

Project Risk Register_x000a_<Project Name>

WAH4Hospital (WAH4H) — Hospital Information System

Risk Identification		Qualitative Rating				Risk Response		
Risk	Risk Category	Probability (1–5)	Impact (1–5)	Risk Score (P×I)	Risk Ranking	Risk Response / Mitigation	Trigger	Risk Owner
R-01 — Proxmox Server Hardware Failure Critical failure of the bare-metal Proxmox server renders the WAH production hosting environment completely unavailable. WAH operates on a single physical server with no confirmed backup node — any hardware failure during deployment, demonstration, or acceptance review blocks all remaining project milestones.	Infrastructure	4	5	20	1	Mitigate. Maintain weekly VM snapshots stored off-server. Conduct monthly hardware health checks (SMART diagnostics, thermal monitoring). Pin all Docker image versions. Document full rebuild procedure in Technical Manual. Coordinate UPS and backup hardware status with Sir Robert Michael Martinez before deployment sprint. <i>Contingency: Restore from latest verified snapshot. If</i>	S.M.A.R.T. disk errors; server fails to POST; unexpected shutdown or reboot; temperature alarms; Sir Robert Michael Martinez reports hardware issues.	Jhon Lloyd Nicolas
R-02 — Cloudflare Tunnel / Network Access Failure Tunnel drops or external URL becomes unreachable, blocking WAH4H access from outside the local network during demonstration or acceptance review.	Infrastructure	2	4	8	2	Mitigate. Configure Cloudflare Tunnel with health checks and alerting. Document fallback local-network access procedure in Technical Manual. Test tunnel stability during sandbox validation.	Tunnel drops during staging test; external URL unreachable; Cloudflare dashboard shows tunnel offline; Fail2Ban blocking legitimate traffic.	Mariyah Vanna Monique C. Chavez
R-03 — Docker Base-Image Incompatibility Container fails to build or run due to architecture mismatch or library incompatibility between development and production environments.	Technical	2	3	6	3	Mitigate. Pin all base image versions in Dockerfile. Test builds on both arm64 and amd64 where applicable. Include build logs in deployment evidence package.	Container exits immediately on start; library linking errors in build logs; architecture mismatch warnings (arm64 vs amd64).	Jhon Lloyd Nicolas
R-04 — Feature Scope Creep Beyond Approved Backlog Informal feature additions expand sprint scope without PM approval, threatening the turnover deadline and overrunning the ₱185,158.50 labor baseline.	Scope	3	3	9	4	Mitigate. No feature development begins without a WBS ID and PM approval. Post-scope-freeze (May 8, 2026), all additions require written authorization from Sir Pey Canlas. PM audits backlog every Friday and flags unauthorized additions.	Sprint velocity drops below baseline; backlog grows mid-sprint; informal feature requests discussed without formal Change Request logging.	Elijah Josh R. Quibin
R-05 — Team Member Availability Disruption A team member becomes unavailable due to academic, personal, or health reasons, stalling assigned deliverables.	Personnel	2	2	4	5	Accept with monitoring. Cross-train team members on critical tasks. Maintain updated task assignments in OpenProject. PM escalates to academic adviser if disruption threatens milestone.	No response within 24 hours during working hours; missed two consecutive standups; assigned tasks stall without update.	Elijah Josh R. Quibin
R-06 — Schedule Overrun — Turnover Deadline at Risk The May–June 2026 turnover deadline is fixed by graduation timeline. Post-UAT defect resolution, documentation completion, and deployment are pending with no confirmed milestone dates.	Schedule	4	5	20	6	Mitigate. Immediate internal team realignment — confirmed milestone dates communicated to Sir Pey Canlas at next WAH touchpoint. Parallel-track execution (UAT fixes + documentation + deployment planning simultaneously). Feature freeze May 8, 2026. <i>Contingency: Negotiate minimum viable turnover scope with Sir Pey Canlas and Sir Robert Michael Martinez; working</i>	No confirmed milestone dates agreed with Sir Robert Michael Martinez; post-UAT fixes consuming more than one sprint; documentation not started by mid-May; no confirmed turnover date by May 30, 2026.	Elijah Josh R. Quibin
R-07 — Lack of Team Communication and Coordination Team members miss standups, blockers go unresolved for 48+ hours, or conflicting implementations are discovered during integration.	Personnel	2	3	6	7	Mitigate. Daily standups via Discord/MS Teams. All blockers logged in OpenProject within 24 hours. PM resolves coordination conflicts at weekly check-in.	Team member misses two consecutive standups; blocker unresolved for 48+ hours; conflicting implementations found during integration.	Elijah Josh R. Quibin
R-08 — Insufficient or Failing Local Development Hardware Development machine unable to run Docker or build containers reliably, blocking sprint progress.	Infrastructure	2	3	6	8	Mitigate. Identify alternate development machine before critical sprint. Use GitHub Codespaces or shared sandbox as fallback. Document hardware requirements in onboarding notes.	Machine unable to run Docker; build times exceed 3× baseline; BSOD/crash reports; insufficient disk space warnings during container operations.	Jhon Lloyd Nicolas

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R-09 — Unplanned Change of Project Direction or Scope Adviser feedback, technology swap, or scope-altering requirement raised without a Change Request, disrupting WBS baseline.	Scope	2	2	4	9	Mitigate. All scope changes require Change Request and PM approval. PM monitors adviser feedback for implied architectural changes. WBS serves as controlling scope reference.	Adviser feedback implies architectural change; technology stack swap proposed informally; scope-altering requirement raised in chat without a Change Request.	Elijah Josh R. Quibin
R-10 — Third-Party API or Service Dependency Failure External API or service dependency returns unexpected errors or announces breaking changes during development or demonstration.	Technical	2	3	6	10	Mitigate. Document all external API dependencies. Maintain fallback mock data for demo. Monitor service status pages. Pin API versions where possible.	API returns unexpected errors in staging; library announces breaking changes; external service status page shows degraded or outage.	Jhon Lloyd Nicolas
R-11 — Data Loss or Corruption in Development Database Migration script fails mid-execution or automated backup job reports failure, causing data loss in the development or staging environment.	Technical	1	4	4	11	Mitigate. Automated daily database backups. Migration scripts tested in sandbox before applying to staging. Backup verification included in deployment checklist.	Migration script fails mid-execution; database refuses to start after an update; automated backup job reports failure.	Jhon Lloyd Nicolas
R-12 — UAT Defects Requiring Significant Post-UAT Rework HIGH-severity defects from UAT feedback (CR-002, May 18, 2026) remain unresolved, blocking Ms. Anna Katrina Yturralde's acceptance sign-off and delaying formal turnover.	Quality / Schedule	3	4	12	12	Mitigate. Triage all open defects immediately by severity. Lead Developer owns all technical fixes. Phased approach: fix → re-test → partial sign-off per module where possible. Escalate blocking defects to Sir Robert Michael Martinez for technical guidance. Contingency: If rework volume threatens deadline, negotiate with Sir Pey Canlas to defer non-critical defects to post.	HIGH-severity defects unresolved after two weeks; re-testing not initiated within one week of fix deployment; Ms. Anna Katrina Yturralde sign-off blocked by open defects.	Elijah Josh R. Quibin
R-13 — Knowledge Gap or Skill Deficiency in Critical Technology Task estimated at 2 days takes 5+ days due to team member's unfamiliarity with a critical technology stack component.	Personnel	2	3	6	13	Mitigate. Identify skill gaps at sprint planning. Assign pairing or external reference materials for complex tasks. PM escalates unresolved technical blockers to Lead Developer.	Task estimated at 2 days takes 5+ days; team member flags a blocker with no proposed solution; multiple failed integration attempts on the same task.	Elijah Josh R. Quibin
R-14 — Security Vulnerability in Deployed Application Dependency audit flags a known CVE or credentials are found in version control history, exposing the system to unauthorized access.	Security	2	4	8	14	Mitigate. Run dependency audit before deployment. Enforce .gitignore for all secrets and .env files. Cloudflare Tunnel and Fail2Ban configured before production deployment.	Dependency audit flags known CVE; credentials found in version control history; unauthorized access attempt logged by Fail2Ban.	Jhon Lloyd Nicolas
R-15 — PH Core FHIR R4 Compliance Gaps Discovered Late Known compliance gaps (Billing module has no PHCore R4 profile alignment; Discharge and Pharmacy Inventory require mapping decisions) require schema migrations that cascade across all dependent modules.	Compliance	3	4	12	15	Mitigate. Complete PH Core FHIR R4 Compliance Matrix (WBS 1.2.6) as hard prerequisite before further module feature work. Billing module is immediate priority. QA validates all FHIR resource mappings against Compliance Matrix before each module is submitted for UAT sign-off. Contingency: Scope compliance fixes to minimum set required for UAT sign-off. Document remaining gaps in	Schema review reveals unmapped FHIR fields in Billing; Procedure_Performer duplication unresolved; Compliance Matrix (WBS 1.2.6) not completed before feature freeze.	Jhon Lloyd Nicolas
R-16 — WAH4P / WAH4PC Interoperability Pending WAH Technical Specification WBS 1.6 integration sprint cannot begin without the technical specification from Sir Robert Michael Martinez defining sync events, data fields, and triggers.	External	4	3	12	16	Mitigate. Immediate escalation to Sir Robert Michael Martinez. Flag as HIGH open item at next WAH touchpoint. Integration sprint (WBS 1.6.2 and 1.6.4) must not begin without confirmed specification. Contingency: If spec remains undefined at sprint start, develop stub sync layer with documented assumptions. Present to Sir Robert Michael Martinez for confirmation and Mitigate. Flag as WAH prerequisite at next meeting. Confirm accreditation and registration status with Sir Pey Canlas. Scope eClaims to be functionally testable with mock data pending accreditation.	No technical spec received from Sir Robert Michael Martinez by integration sprint start; WBS 1.6.1 and 1.6.3 remain blank; turnover approaching with WAH4P/WAH4PC sync unvalidated.	Jhon Lloyd Nicolas / Sir Robert Michael Martinez
R-17 — PhilHealth Accreditation / NHFR Registration Not Completed by WAH PhilHealth accreditation and NHFR registration are WAH prerequisites outside Team Pseudoers' scope. Without accreditation, eClaims batch submission cannot be tested against live PhilHealth systems.	External	3	4	12	17	Contingency: Deliver eClaims as functionally complete with simulated PhilHealth submission workflow. Document in Technical Manual that live submission requires WAH to	No confirmation from Sir Pey Canlas of WAH's PhilHealth accreditation status; NHFR registration status unknown at eClaims sprint start.	Elijah Josh R. Quibin

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R-18 — Proxmox Production VM Not Provisioned on Time by WAH WBS 1.10.1 Production VM Provisioning is the responsibility of Sir Robert Michael Martinez. If not provisioned before the deployment sprint, WBS 1.10.2 through 1.10.4 are blocked.	External	3	4	12	18	Mitigate. Confirm Proxmox provisioning timeline with Sir Robert Michael Martinez and Sir Pey Canlas at next touchpoint. Include this as a tracked hard dependency milestone. Maintain fully functional sandbox as fallback for demonstrations. Contingency: If Production VM not ready at turnover, deliver system on sandbox with complete Technical Manual.	No confirmation from Sir Robert Michael Martinez on VM readiness; deployment sprint starting without a provisioned production environment; Cloudflare Tunnel not yet configured on WAH infrastructure.	Elijah Josh R. Quibin
R-19 — Incomplete Stakeholder Contact Information Contact details for Ms. Anna Katrina Yturralde, Ma'am Nicky Balderosa, Sir Robert Michael Martinez, Sir Pey Canlas, and Ma'am Seann Biag are TBD in the Stakeholder Register. Missing contacts block formal communication, escalation, and acceptance sign-off coordination.	Communications	4	3	12	19	Mitigate. Collect all contact details at next WAH touchpoint. Ma'am Nicky Balderosa is primary day-to-day POC — her details must be collected first. Contingency: Route all communications through Sir Robert Michael Martinez (montemyk@gmail.com / Messenger — only confirmed contact) until direct contacts are established.	Stakeholder register still has TBD contacts two weeks into execution; unable to reach Ms. Anna Katrina Yturralde; escalation attempt blocked by missing contact.	Elijah Josh R. Quibin
R-20 — Ms. Anna Katrina Yturralde Unavailable for Final Acceptance Review Ms. Anna Katrina Yturralde is the required acceptance authority. Sir Pey Canlas's final sign-off cannot proceed without her prior approval. If unavailable within the May–June 2026 window, formal project closure is blocked regardless of system readiness.	External	3	5	15	20	Mitigate. Obtain Ms. Anna Katrina Yturralde's contact details immediately. Schedule acceptance review at least three weeks before target turnover date. Communicate schedule constraints to Sir Pey Canlas proactively. Contingency: Escalate to Sir Pey Canlas to designate interim acceptance authority or extend the sign-off window. Compile complete system readiness evidence package (LIAT results).	No confirmed availability from Ms. Anna Katrina Yturralde by mid-May; contact details still TBD; acceptance review not scheduled three weeks before target turnover date.	Elijah Josh R. Quibin
Key Terms								
Risk: The risk stated in a complete sentence which states the cause of the risk, the risk, and the effect that the risk causes to the project.								
Risk Category: Categorization of risks by area of project affected (Infrastructure, Technical, Scope, Personnel, Schedule, Quality/Schedule, Security, Compliance, External, Communications).								
Probability: The likelihood that a risk will occur. Scored 1–5 on the 5×5 matrix: 1=Rare, 2=Unlikely, 3=Moderate, 4=Likely, 5=Almost Certain.								
Impact: The severity of effect on project objectives if the risk materializes. Scored 1–5 on the 5×5 matrix: 1=Insignificant, 2=Minor, 3=Significant, 4=Major, 5=Severe.								
Risk Score: Determined by multiplying Probability × Impact. Scale from 1 to 25.								
Risk Ranking: Sequential risk ID number for tracking purposes.								
Risk Response: Proactive mitigation strategy (taken before event occurs) and pre-approved contingency plan (activated when risk materializes).								
Trigger: Observable indicator or early warning signal that a risk is about to materialize.								

Risk Owner: Team member responsible for monitoring the risk and executing its response plan.

Priority Zones (5×5 RMP v2 Matrix): EXTREME ≥20 (Red) | VERY HIGH 15–19 (Orange) | HIGH 10–14 (Yellow) | MEDIUM 6–9 (Green) | LOW 3–5 (Blue) | VERY LOW 1–2.